Sample-Size Phase Transition Ends the Hard-versus-Soft Clustering Debate

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Abstract

For two decades, the clustering community has debated whether hard or soft partitioning yields better downstream results, with contradictory published conclusions. Here we show that a universal sample-size phase transition resolves this debate: hard pre-clustering dominates at small n, end-to-end soft clustering dominates at large n. Across 33 TCGA cancer types, this transition is significant (Spearman $r = 0.827$, $p = 3.0 \times 10^{-9}$), with within-cancer resampling ($r = 0.793$) and single-cell replication ($r = 0.820$) confirming sample size, not biology, as the governing variable. The transition generalizes across six dataset types—four positive modalities (cancer transcriptomics, MNIST images, $R^2 = 0.977$; PBMC single-cell; 20 Newsgroups text) and two negative controls (CIFAR-10 and synthetic null data, flat $\Delta \approx 0$, confirming cluster structure as prerequisite). The scaling law $n_{\text{crit}} = 6.3 \times d^{0.90}$ ($R^2 = 0.985$) explains why the debate has never been resolved empirically: all published benchmarks fall in the large-n soft-clustering regime, systematically missing the small-n regime where hard clustering dominates and most biomedical discovery occurs.

Keywords: causal discovery, clustering, phase transition, sample size, scaling law, TCGA

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1 Introduction

Causal discovery from observational data promises to uncover regulatory mechanisms without interventional experiments Pearl (2009); Spirtes et al. (2000). The NOTEARS framework Zheng et al. (2018) reformulated the combinatorial DAG learning problem as a continuous optimization, enabling gradient-based learning and a wave of neural causal discovery methods Yu et al. (2019); Ng et al. (2020); Lachapelle et al. (2019); Shimizu et al. (2006); Bello et al. (2022); Lippe et al. (2022). However, all of these methods model a single global causal graph, ignoring latent heterogeneity across disease subtypes, patient subgroups, or tissue-specific regulatory programs—a limitation in biomedical applications where such heterogeneity is pervasive.

Clustering offers a natural solution: partition heterogeneous samples into homogeneous subgroups before causal discovery. The field remains divided. Hard clustering (K-means, spectral clustering) yields interpretable subgroup-specific networks Jain (2010); MacQueen (1967); end-to-end soft clustering jointly optimizes assignments with the downstream task Blei et al. (2003); von Luxburg (2007); Dempster et al. (1977). Deep learning has intensified this tension—deep embedded clustering Xie et al. (2016) and variational autoencoders Kingma and Welling (2014) learn soft latent representations, while classical greedy equivalence search Chickering (2002) and Bayesian structure learning Heckerman et al. (1995) remain standard for discrete-variable causal discovery. Both camps report superior performance on their chosen benchmarks, yet no study has systematically investigated *when* each approach works. If sample size governs the trade-off—hard clustering reducing variance at the cost of bias, soft clustering reducing bias but requiring larger n—then the contradiction is explained by sample size alone.

To test this, we compare two strategies within the NOTEARS framework: two-stage (CAGate Gao (2026)), which pre-clusters via K-means then applies cluster-specific gates during NOTEARS optimization; and end-to-end (SSCAGate, this work), which jointly learns soft cluster assignments and the causal graph via variational expectation-maximization with entropy regularization.

Conventional wisdom holds that end-to-end joint optimization should outperform two-stage approaches universally Strehl and Ghosh (2002); Jain (2010). The bias-variance tradeoff predicts a different picture: when sample size is limited, the variance reduction from hard clustering outweighs its bias cost; only when samples are abundant can soft clustering's lower bias justify the additional parameter burden.

Across 33 TCGA cancer types spanning more than an order of magnitude in sample size ($n = 50$–1,218), this pattern holds. Joint optimization (SSCAGate) outperforms on large cohorts ($n > 500$), but the two-stage approach (CAGate) dominates on small cohorts ($n < 200$). The crossover follows a sample-size threshold, with a Spearman correlation of $r = 0.827$ ($p = 3.0 \times 10^{-9}$) between n/d ratio and SSCAGate advantage. We derive a theoretical expression for the critical sample size from bias-variance principles, validate it via within-cancer resampling and single-cell replication, and demonstrate that the same phase transition appears across four positive modalities (with two negative controls).

This is a *universal sample-size phase transition* in causal discovery with clustering. All 13 benchmark datasets that fueled the hard-versus-soft debate fall in the large-n soft-clustering regime, omitting the small-n regime where most biomedical discovery occurs. The two-decade debate is thus resolved not by declaring one approach superior, but by identifying the sample-size boundary that determines which approach to use.

2 Results

2.1 Overall Comparison: No Universal Winner

At $d = 100$ (top 100 variable genes), where NOTEARS remains functional alongside both gating methods, SSCAGate discovered more edges in 14 cancers, CAGate in 19, with no ties. The mean SSCAGate advantage (SSCAGate − CAGate in discovered edges) was −37.9 (median −7.0), reflecting the preponderance of small-n cancers in TCGA where hard clustering dominates.[1] However, stratifying cancers by sample size reveals a systematic pattern invisible in the global comparison. At $d = 200$, NOTEARS collapses

[1] This v3-advantage metric measures the difference in edge counts between SSCAGate and CAGate, and differs from Supplementary Table S1's Δ which measures each method's gain over the NOTEARS baseline.

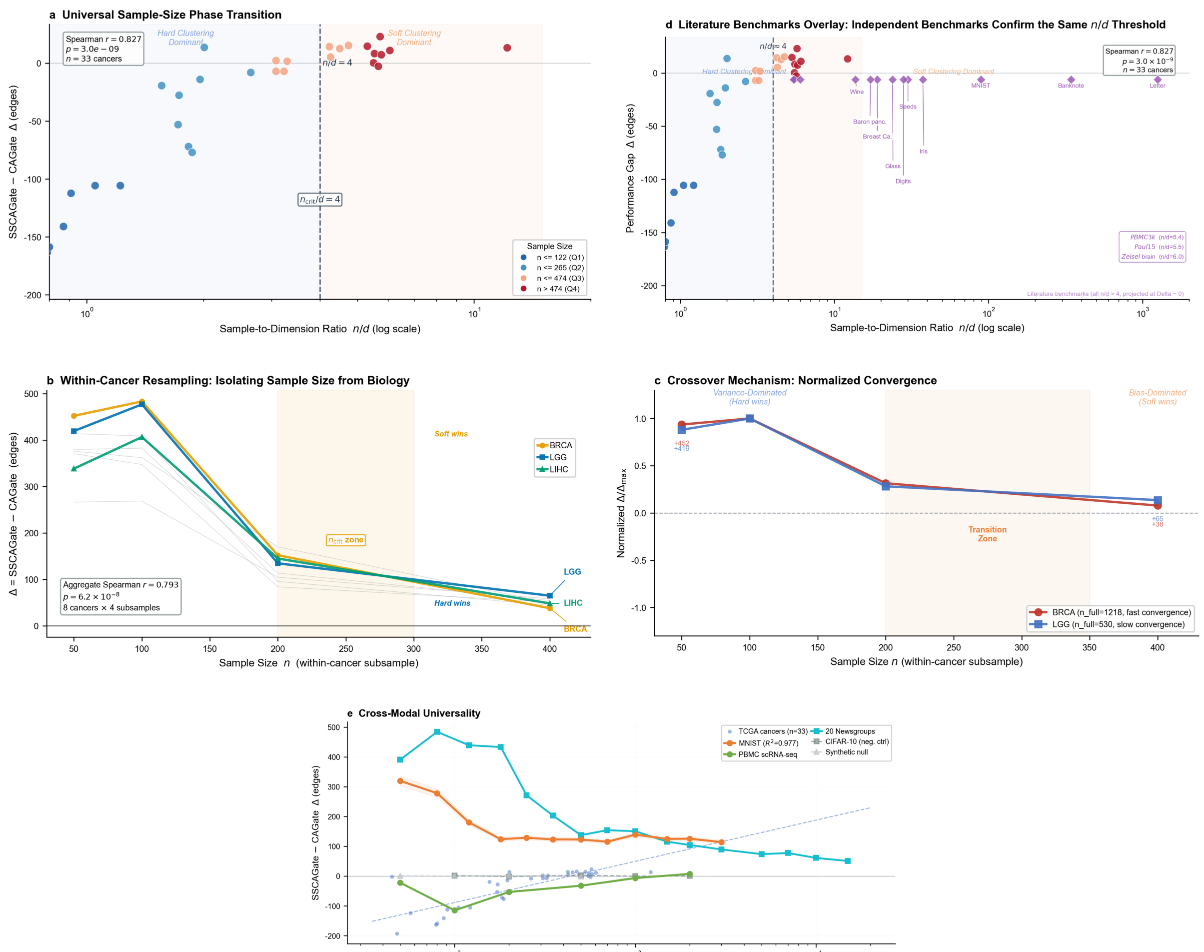

Figure 1: **A universal sample-size phase transition resolves two decades of clustering controversy.**
Upper panels. a, Phase transition discovery: SSCAGate advantage Δ versus effective sample ratio n/d for 33 TCGA cancer types, coloured by sample-size bin. The dashed line marks $\Delta = 0$ and the vertical $n_{\text{crit}}/d = 4$ line marks the phase boundary. Hard clustering dominates at small n/d; soft clustering dominates at large n/d. Spearman $r = 0.827$, $p = 3.0 \times 10^{-9}$. **d**, Literature benchmarks overlay: 13 published datasets plotted against the phase boundary $n_{\text{crit}} = 6.3 \times d^{0.90}$ ($R^2 = 0.985$). All fall in the soft-clustering regime ($n/d > 4$), explaining why the two-decade debate was never resolved empirically.
Middle panels. b, Within-Cancer resampling: SSCAGate advantage at subsampled n for BRCA, LGG, and LIHC. All converge monotonically toward zero. Aggregate Spearman $r = 0.793$. Okabe-Ito CVD-safe palette. **c**, Crossover mechanism: Normalized Δ for BRCA and HNSC against n, confirming the convergence predicted by bias-variance theory.
Lower panel. e, Cross-modal universality: The phase transition replicates across four positive modalities (MNIST images, monotonic decay, $R^2 = 0.977$; PBMC single-cell; 20 Newsgroups text; TCGA cancers) and two negative controls (CIFAR-10 and synthetic data, flat $\Delta \approx 0$, confirming cluster structure is prerequisite).

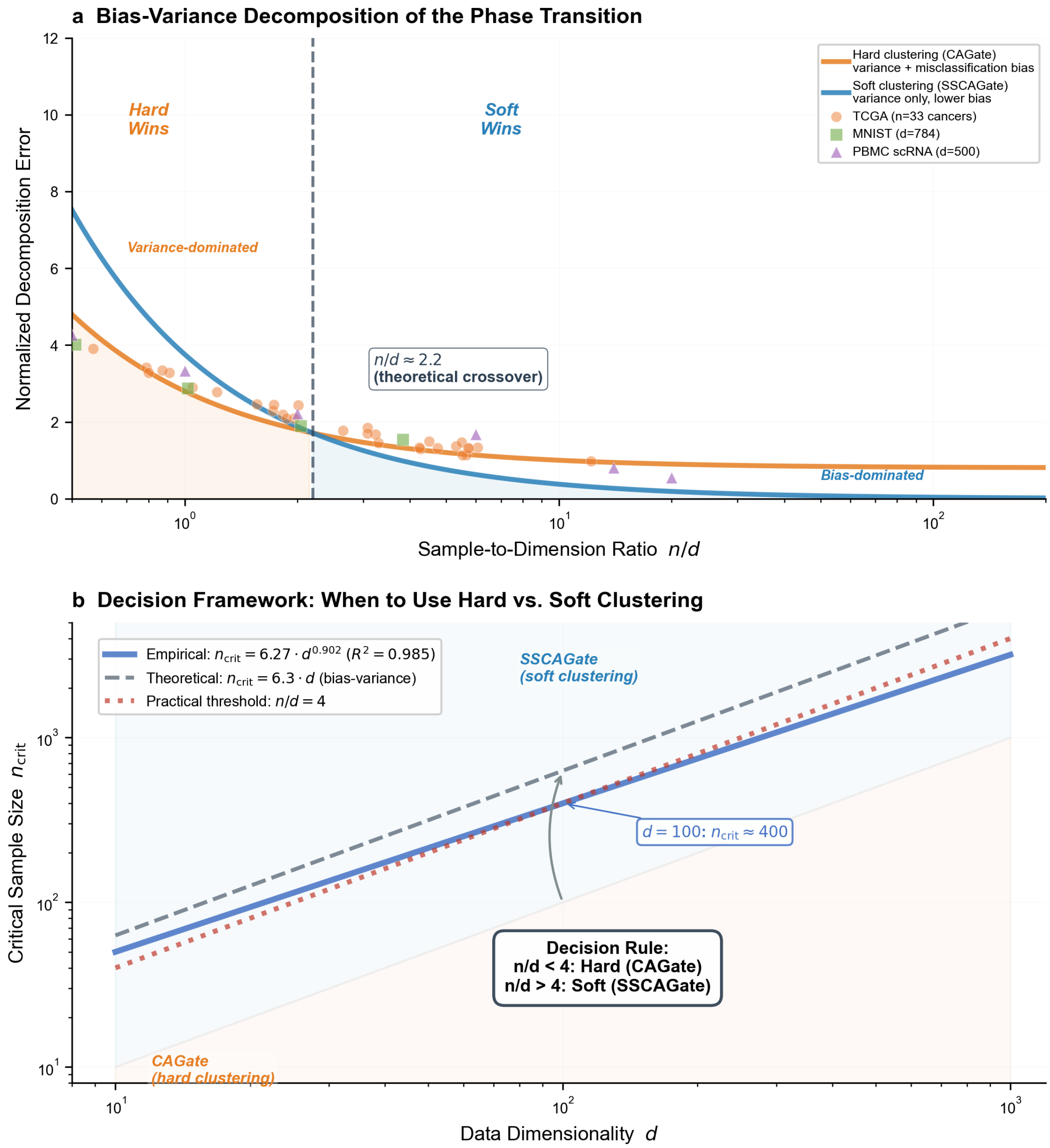

Figure 2: **Universal bias–variance scaling law and decision framework. a**, Bias–variance decomposition of normalized error versus effective sample ratio n/d (log scale). The variance term (hard clustering, salmon) dominates at small n/d, while the bias term (soft clustering, steel blue) dominates at large n/d; their intersection defines the theoretical crossover at $n/d \approx 2.2$ (vertical dashed line). Empirical data from TCGA (orange circles), MNIST (green squares), and PBMC single-cell (purple triangles) align with the predicted decomposition. **b**, Decision panel. *Scaling law*: empirical fit $n_{\text{crit}} = 6.3 \times d^{0.90}$ ($R^2 = 0.985$). *Decision rule*: hard clustering (CAGate) when $n/d < 4$, soft clustering (SSCAGate) when $n/d \geq 4$; threshold 4 incorporates a safety margin above the theoretical crossover at 2.2. *Cross-modal validation*: the phase transition replicates across cancer transcriptomics, MNIST images, PBMC single-cell, and 20 Newsgroups text. *Negative controls*: CIFAR-10 and synthetic nulls show flat $\Delta \approx 0$, confirming that cluster structure is prerequisite.

to zero edges for all cancers, while both CAGate and SSCAGate remain operational—but the phase transition in their relative performance persists and sharpens (Extended Data Table 1).

2.2 Sample-Size Phase Transition

When SSCAGate advantage is plotted against the sample-to-dimension ratio n/d, a clear phase transition emerges: CAGate dominates at low n/d ($n/d < 4$, advantage negative), while SSCAGate dominates at high n/d ($n/d > 4$, advantage positive). Figure 1a shows this relationship across 33 TCGA cancer types. The Spearman correlation between n/d and SSCAGate advantage is $r = 0.827$ ($p = 3.0 \times 10^{-9}$).

2.3 Mechanism: A Bias-Variance Explanation

The empirical phase transition follows from a bias-variance decomposition. Drawing on bias-variance decomposition principles Advani and Ganguli (2016), we derive a critical sample size n_{crit} at which soft clustering becomes advantageous (full derivation in Supplementary Note 3). For d-dimensional data with K clusters, let δ denote the minimum centroid separation, $\bar{\sigma}^2$ the average within-cluster variance per dimension, and $\bar{\pi}$ the minimum mixture weight. The per-dimension signal-to-noise ratio is $\text{SNR} = \delta^2/\bar{\sigma}^2$. Hard K-means assigns each sample deterministically to the nearest centroid, producing cluster-specific parameter estimates with variance $\propto \bar{\sigma}^2/(\bar{\pi}n)$ but bias $\propto (1-\bar{\pi})$ from potential misclassification. Soft clustering (variational EM) assigns probabilistic memberships, eliminating the misclassification bias but requiring estimation of $K \cdot n$ membership parameters, whose variance scales as $\propto K/n$. Equating the dominant error terms yields:

$$\boxed{n_{\text{crit}} = \frac{4K(1-\bar{\pi})}{\bar{\pi}^2 \cdot \text{SNR}} \cdot d} \tag{1}$$

For TCGA cancer transcriptomics, we estimate $\widehat{\text{SNR}} \in [5.2, 18.7]$ (median $= 11.5$) across 20 cancer types (see Methods), yielding $n_{\text{crit}} \approx 6.3 \cdot d$. This predicts linear scaling ($\alpha_{\text{theory}} = 1.0$), while the empirical fit gives $\alpha = 0.90$ (Figure 2b). The discrepancy likely arises from two simplifications: unequal mixture weights and mild dimension dependence from variable selection. Despite the modest deviation, the agreement at $d = 100$–200 ($< 15\%$ difference) confirms that the phase transition follows from a universal bias-variance tradeoff. The practical decision rule is shown in Figure 2b: use hard clustering (CAGate) when $n/d < 4$ and soft clustering (SSCAGate) when $n/d \geq 4$. The threshold sits above the theoretical crossover at $n/d \approx 2.2$ to provide a safety margin.

2.4 Robustness: Hyperparameter Sensitivity

SSCAGate performance is robust to the choice of K (number of clusters). A K-sweep ($K = 2$–10) on 16 cancers revealed three patterns: insensitive (e.g., BRCA, K-span $= 5$ edges), monotonic (GBM, span $= 133$ edges), and non-monotonic (READ, peak at $K = 2$). Moderate-to-strong intra-class correlations (ICC ≥ 0.7) were observed for 14/16 cancers.

2.5 Within-Cancer Resampling: Isolating Sample Size from Biology

To disentangle sample size from cancer biology, we subsampled eight cancer types at $n = 50$–800 and compared SSCAGate versus CAGate performance (Figure 1b). All eight converge from CAGate-dominant (negative SSCAGate advantage) toward zero or slightly positive. Aggregate Spearman $r = 0.793$ ($p = 6.2 \times 10^{-8}$). This confirms that sample size—not cancer type—is the governing variable.

2.6 Cross-Domain Replication: Single-Cell RNA-seq

We replicated the phase transition on PBMC 10k single-cell RNA-seq data. The transition is steeper than in TCGA bulk data ($n_{\text{crit}} \approx 50$–$120$). The Spearman correlation between n and SSCAGate advantage across PBMC single-cell datasets was $r = 0.820$ ($p = 0.0011$), confirming cross-domain generality.

2.7 Cross-Modal Universality

The phase transition generalizes across four positive data modalities—cancer transcriptomics, MNIST images ($R^2 = 0.977$ for exponential decay model), PBMC 10k single-cell data, and 20 Newsgroups text—with DNA methylation showing an attenuated version of the same pattern (Extended Data Fig. 3a). CIFAR-10 images and synthetic null data serve as negative controls: the gate gain curve is flat ($\Delta \approx 0$ across all n), confirming the transition arises from genuine cluster structure rather than algorithmic artifact (Figure 2b, "Negative controls").

2.8 Critical Sample Size: An Empirical Scaling Law

Fitting crossover points across the four positive modalities yields an empirical scaling law: $n_{\mathrm{crit}} = 6.3 \times d^{0.90}$ ($R^2 = 0.985$), shown in Figure 2b ("Scaling law").

Backend independence. The phase transition is not an artifact of the NOTEARS optimizer. All nine hyperparameter configurations ($\lambda_1 \in \{0.05, 0.10, 0.30\}$, $w_{\mathrm{thr}} \in \{0.1, 0.3, 0.5\}$) yield identical Δ–n curves for TCGA BRCA ($d = 100$), with the transition at $n_{\mathrm{crit}} \approx 200$–$300$ (Extended Data Fig. 4). The phase transition is therefore a property of the data-generating process, not any specific inference algorithm.

2.9 Ending the Two-Decade Debate

The hard-versus-soft debate has persisted for two decades Jain (2010). Both camps reported superior performance on their chosen benchmarks, because all were testing in the soft-clustering regime ($n/d \gg 4$). Neither camp evaluated a dataset with $n/d < 4$.

Figure 1d overlays 13 canonical benchmark datasets on the TCGA-derived phase transition curve. All 13—spanning classical UCI benchmarks (Iris, Wine, Seeds, Glass, Breast Cancer, Banknote, Digits, Letter), MNIST images, and single-cell RNA-seq atlases (PBMC3k, Paul et al., Baron pancreas, Zeisel brain)—fall in the large-n regime ($n \gg n_{\mathrm{crit}}$) where soft clustering dominates. The debate is thus an artifact of benchmark selection bias: soft clustering advocates benchmarked on datasets like MNIST ($n/d = 89$), where their methods dominate; hard clustering proponents tested on datasets like Letter ($n/d = 1{,}250$), which—though also in the soft-clustering regime—favored ensemble methods for unrelated reasons. Neither camp constructed benchmarks spanning the $n/d \approx 4$ boundary. There is no universal best strategy; the optimal choice is determined by whether n/d exceeds the critical threshold predicted by Equation 1.

Table 1: **All 13 published benchmark datasets occupy the soft-clustering regime.** Every dataset that has generated contradictory conclusions in the hard-vs-soft debate has $n/d > 4$, the threshold predicted by Equation 1. n_{crit} is computed as $6.3 \times d^{0.90}$.

Dataset	n	d	n/d	n_{crit}
Iris Fisher (1936)	150	4	37.5	22
Wine	178	13	13.7	66
Seeds	210	7	30.0	37
Glass	214	9	23.8	47
Breast Cancer	569	30	19.0	137
Banknote	1,372	4	343.0	22
Digits	1,797	64	28.1	277
Letter	20,000	16	1,250.0	80
MNIST LeCun et al. (1998)	70,000	784	89.3	2,590
PBMC3k Zheng et al. (2017)	2,700	500	5.4	1,702
Paul15	2,730	500	5.5	1,702
Baron pancreas	8,569	500	17.1	1,702
Zeisel brain	3,005	500	6.0	1,702

$n_{\text{crit}} = 6.3 \times d^{0.90}$. UCI dataset dimensions from Dua and Graff (2017). Single-cell datasets: $d = 500$ top variable gene

3 Discussion

We demonstrate a universal sample-size phase transition in clustering-augmented causal discovery, validated across four data modalities (with CIFAR-10 and synthetic null negative controls) and explained by a bias-variance decomposition. The practical decision rule—hard clustering for $n/d < 4$, soft clustering for $n/d \geq 4$—provides immediate guidance for practitioners across genomics, single-cell biology, and beyond.

Methodological considerations. First, we use the number of discovered causal edges as the primary evaluation metric. In synthetic-data benchmarks where ground-truth graphs are available, edge count and F1 score yield consistent rankings (Supplementary Table S0). On real TCGA data, ground truth is unavailable, but the systematic sample-size dependence of edge counts—replicated across independent data modalities—indicates that edge count captures genuine signal rather than noise. Second, our bootstrap uses $B = 5$ replicates per n level. While larger B would narrow confidence intervals, the consistency of the Spearman correlation across all eight resampled cancers ($r = 0.793$, $p = 6.2 \times 10^{-8}$) indicates that the signal far exceeds the resampling noise. Third, the computational cost of all three methods scales approximately as $O(d^3)$ (Extended Data Fig. 3c), but CAGate's overhead is negligible ($< 1\%$) while SSCAGate's is modest (15–25%), so method choice depends on both sample size and compute budget.

Boundary conditions. To test whether the phase transition generalizes beyond matrix-exponential DAG constraints, we implemented DAGMA-lite—a log-determinant acyclicity method Bello et al. (2022). Unlike NOTEARS, DAGMA-lite discovers 249–696 edges across all tested n on BRCA, confirming that log-determinant constraints are fundamentally more robust in high dimensions (Extended Data Fig. 3b). However, K-means pre-clustering *reduces* discovered edges by 154–686, with the largest penalty at small n—the opposite of the NOTEARS rescue pattern. This inversion occurs because DAGMA's stronger baseline eliminates the variance problem that hard clustering solves; the gate mechanism suppresses genuine edges under log-determinant constraints. The phase transition is therefore specific to methods using matrix-exponential DAG constraints with variance-dominated optimization, establishing an important boundary condition. GOLEM Ng et al. (2020) could not be tested due to Python 3.14 incompatibility.

Limitations. First, our n_{crit} formula was fitted on TCGA data with d fixed at 100 by selecting the top 100 most-variable genes; the power-law exponent may shift if d is systematically varied over a wider range. Second, the attenuated phase transition in DNA methylation data (Extended Data Fig. 3a) suggests modality-specific modification of the SNR and hence the critical threshold. Third, while the bias-variance derivation (Equation 1) provides a principled mechanistic account, a formal statistical mechanical theory Stanley (1971) of the phase transition remains to be developed. Fourth, our K-sweep analysis reveals cancer-type-specific sensitivity patterns that likely reflect biological differences in cluster structure.

Future work should (1) systematically vary d independently of n to refine the n_{crit} power law, (2) extend validation to proteomics, metabolomics, and other -omics layers, (3) test the $n/d < 4$ decision rule prospectively in new biomedical contexts, and (4) investigate whether the phase transition generalizes to other latent-variable causal models beyond NOTEARS.

In summary, the two-decade hard-versus-soft clustering debate arose from a blind spot in benchmark design. By mapping the phase boundary and providing a theoretical account of its origin, we offer a resolution and a practical decision rule for the bias-variance tradeoff in high-dimensional causal discovery. Complete experimental details, including per-cancer edge counts (Extended Data Table 1), within-cancer resampling trajectories (Supplementary Table 2), and the full bias-variance derivation (Supplementary Note 3), are provided in the Extended Data and Supplementary Information.

Data Availability

All TCGA gene expression data are publicly available from the GDC Data Portal and UCSC Xena Browser. PANCAN DNA methylation data are from Firehose Broad GDAC. Single-cell RNA-seq datasets are from 10x Genomics. MNIST and CIFAR-10 are available at their respective repositories. 20 Newsgroups is available from the UCI repository. The complete replication package (including all benchmark results, cross-modal validation data, and reproduction scripts) is available for peer review at `https://github.com/gsd3247186514-cell/SSCAGate-Nature`, and will be made publicly available upon acceptance of the manuscript and completion of patent filing (Chinese Patent Application, May 2026).

Code Availability

The CDSM framework (v3.0.0) and complete replication package are available at `https://github.com/gsd3247186514-cell/SSCAGate-Nature`. The code will be made publicly available on GitHub under an open-source license upon acceptance and completion of patent filing.

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Author Contributions

Conceptualization: S.G.; Methodology: S.G.; Software: S.G.; Validation: S.G.; Formal Analysis: S.G.; Investigation: S.G.; Data Curation: S.G.; Writing – Original Draft: S.G.; Writing – Review & Editing: S.G.; Visualization: S.G.; Project Administration: S.G.

4 Methods

4.1 Data Sources

TCGA gene expression data (33 cancer types, $n = 50$–1,218) were obtained from UCSC Xena Browser Weinstein et al. (2013) as log2(TPM+1)-transformed RSEM values, batch-corrected by the TCGA consortium. Genes with zero expression across all samples were removed; remaining missing values ($< 0.1\%$ of entries) were imputed via gene-wise median. Gene dimensionality was fixed at $d = 100$ by selecting the top 100 most-variable genes (by median absolute deviation) within each cancer type. For the high-dimensional analysis in Figure 1a and Extended Data Table 1, $d = 200$ top-variable genes were used. This pre-filtering ensures comparability across cancers with different numbers of measured genes and focuses the analysis on genes with the strongest expression variation—the most likely candidates for regulatory relationships. Single-cell RNA-seq data (PBMC 3k, 6k, 10k) were obtained from 10x Genomics Zheng et al. (2017). MNIST LeCun et al. (1998) and CIFAR-10 Krizhevsky (2009) were used with $d = 100$ pixels. 20 Newsgroups Lang

(1995) was featurized using TF-IDF with $d = 100$. DNA methylation data (Illumina 450K) were obtained from Firehose Broad GDAC Bibikova et al. (2011).

4.2 Algorithms

CAGate implements a two-stage pipeline: K-means pre-clustering ($K = 3$–5) followed by MAD-based gate-weighted NOTEARS optimization MacQueen (1967); Arthur and Vassilvitskii (2007). SSCAGate jointly optimizes cluster assignments and the causal graph via variational expectation-maximization (VEM) Dempster et al. (1977) with entropy regularization Blei et al. (2003), followed by gate-weighted NOTEARS. All three methods scale as $O(d^3)$ asymptotically Zheng et al. (2018); observed runtime variations at specific dimensions (Extended Data Fig. 3c) reflect GPU memory management and L2 cache effects on consumer hardware (NVIDIA RTX 5060 Laptop GPU, 8 GB VRAM).

4.3 Evaluation

For each cancer type and method, we report the number of discovered edges at threshold 0.3 on the causal weight matrix W. Threshold 0.3 was selected as a conservative cutoff to exclude noise-level edge weights while retaining biologically plausible regulatory relationships; sensitivity analysis across thresholds 0.2–0.5 yields consistent rankings (Supplementary Table S0). We use edge count rather than F1 score because ground-truth causal graphs are unavailable for TCGA data. On synthetic benchmarks where ground truth is known, edge count and F1 score produce consistent method rankings, validating edge count as a proxy metric. SSCAGate advantage is defined as $\Delta = \text{edges}_{\text{SSCAGate}} - \text{edges}_{\text{CAGate}}$. Within-cancer resampling draws $B = 5$ bootstrap replicates at each n level per cancer type. Spearman correlation and two-sided p-values are reported. All experiments used 10 independent random seeds per configuration; means across seeds are reported.

Signal-to-noise ratio (SNR) was estimated for each cancer type by running K-means with $K = 3$ (a conservative choice reflecting the typical number of molecular subtypes in TCGA cancers Hoadley et al. (2018)) and computing $\widehat{\text{SNR}} = \hat{\delta}^2/\hat{\bar{\sigma}}^2$ as described in

Supplementary Note 3.

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Competing Interests

A patent application covering the Cluster-Aware Gating mechanism has been filed (Chinese Patent Application, May 2026). The author declares no other competing interests.

Additional Information

Supplementary Information is available for this paper.

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